MD. Shamiul Basher Siam

Computer Science & Engineering Student · Software Development · DSA & Algorithms · Agentic AI

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PROFESSIONAL SUMMARY

Computer Science & Engineering student at KUET, focused on software engineering, data structures & algorithms, and agentic AI. Builds complete C++, Python, and Verilog/FPGA projects — from a polymorphic water-grid system to an autonomous AI agent and a genomic sequence search engine.

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog HDL, Assembly language

Core CS & Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, STL, Templates, Exception Handling, Operator Overloading

AI & Agentic Systems: Agentic AI, Tool Calling, Memory Retrieval, Planning Systems, Plugin Architecture, Gemini API, SQLite

Hardware & Digital Logic: Xilinx Vivado, Basys 3 FPGA, Digital Logic Design, FSM, LFSR

Tools: Git, GitHub, VS Code

SELECTED PROJECTS

HydroSmart – Intelligent Water Grid Management System

2026

C++ · Individual Academic Project · Completed

- Built a modular C++ architecture with an abstract GridModule base class, runtime polymorphism, and inheritance.
- Used STL containers (map, vector), template utilities, and operator overloading for reusable grid operations.
- Added exception-safe file handling with CSV persistence for reliable, real-time grid health analysis.
- Added OTP-based authentication with account-lock protection to secure system access.

FPGA Random Number Guessing Game

2026

Verilog HDL · Basys 3 FPGA · Team Academic Project · Completed

- Built a modular Verilog architecture with an LFSR for pseudo-random number generation on a Basys 3 FPGA.
- Developed an FSM to manage two-player game logic, with switch and push-button input handling.
- Added automatic winner detection with real-time seven-segment and LED feedback, plus reset/replay.
- Collaborated on a team project to deliver a hardware game demonstrating digital logic design.

Agent Prime – Autonomous AI Agent System

In Progress

Python · Gemini API · SQLite · Individual Developer · Expected October 2026

- Designing an agentic AI system with a central Agent Core, SQLite-backed memory, and a planning layer.
- Building a centralized tool registry and plugin architecture for extensible tool calling and Gmail integration.
- Developing an approval-gated execution flow with structured logging for safe, auditable actions.

GenoSearch – Genomic Sequence Search & DNA Pattern Matching Engine

In Progress

C++ · Data Structures & Algorithms · Individual Developer · Expected November 2026

- Developing a C++ genomic search engine using suffix tries, hash tables, and k-mer indexing for DNA/RNA matching.
- Applying rolling hash and dynamic programming for sequence alignment, mutation detection, and similarity ranking.
- Planning a FastAPI, React, and PostgreSQL stack with D3.js to visualize algorithmic search results.

EDUCATION

Khulna University of Engineering & Technology (KUET)

Khulna, Bangladesh

BSc in Computer Science & Engineering (CSE) | Expected Graduation: **2029** | CGPA: **ABOVE 3.30**

ADDITIONAL EXPERIENCE

Online Q&A Tutor — HSC-Level Students

- Explained academic concepts to HSC-level students in one-on-one and group sessions, building communication skills.
- Maintained consistent, independent engagement with students, managing sessions responsibly.